# Borrower Copilot — How to Actually Win This

*This challenge isn't graded on your code. It's graded on whether your rules survive contact with three specific human beings, and on whether you can defend those rules live afterward. Read this once before you touch a keyboard.*

Challenge: Lokta · Borrower Copilot Build Challenge, v1.0, issued 2 Sep 2026 · Time box: 4 days / 12–16 hrs
Comes with: BUILD_PROMPT.md (paste-ready) and borrower_copilot_poc.html (working rules-engine demo)

## 1. What this challenge is actually testing

Strip away the UI and this is a founding-engineer problem, not a coding-test problem: can you take the fuzzy judgement a good branch manager carries in their head — “this one's fine, that one I'd route to a secured product, that one I'd turn away— and turn it into rules explicit enough for a machine to run and honest enough for a stranger to trust with their own money.

Lokta's actual product bet, stated plainly in the brief, is that the borrower's problem is information asymmetry, not access. Anyone with a phone can get *a* loan. Almost no one knows what a *fair* loan looks like for their specific file. That's why RULES.md is read as carefully as the code — the artifact they actually want is the reasoning. The code is just the runtime for it. This is also why they explicitly tell you AI tools are fine to build with, then warn you they'll ask you to change a rule live: they don't care how the reasoning got written down, only whether it's *yours* once it exists.

> **So the job is:** encode judgement as explicit, inspectable rules, and be honest about exactly where that judgement runs out.

## 2. The four sentences that decide your score

Read the scoring table closely and it collapses to four concrete checks, lifted almost verbatim from their own rubric. Nail these for all three personas and you are most of the way to a strong score, independent of how polished the UI is:

- Are the **lender's number and the borrower's number** in O2 correctly different — and can you say why in one sentence?
- Does **“Don't borrow”** actually fire for the profile that should trigger it, or is it a decorative option that never appears?
- Is **Ravi routed to a secured product** (against his shop) instead of priced like an unsecured borrower with no credit history?
- Is the **fair-rate output honest about fees** — an APR, not just the headline rate re-labelled?

Everything in Sections 3–4 below is built to make all four of these true, with numbers you can defend out loud.

## 3. The domain framework

None of these are RBI mandates unless stated — they're judgement calls, which is exactly what RULES.md's “my judgement” column exists for. Change any number you disagree with; keep the *shape* of the reasoning.

### Effective income by employment type

- **Salaried:** declared net salary, taken as-is.
- **Self-employed:** (last ITR annual income ÷ 12) + a haircut on any cash income above that — 40% haircut if unverified, 15% if 6 months of bank statements are shown. Business vintage under 3 years falls back to the informal ceilings below: a young business hasn't proven stability yet.
- **Informal / gig:** midpoint of the stated income range; add a co-earner's income only if they're a formal co-applicant on the loan.

### FOIR-style affordability ceilings (share of effective income available for EMI)

| Segment | Lender-view ceiling | Borrower-safe ceiling |
|---|---|---|
| Salaried, income ≥ Rs 50k/mo | 55% of gross income | 50% of (income – existing EMI – household expenses) — usually the tighter number |
| Salaried, income < Rs 50k/mo | 45% of gross income | same disposable-income method |
| Self-employed, ≥3yr vintage | 45% of effective income | same disposable-income method |
| Informal / gig | 35% of effective income | 25%, ±10 pts by stated emergency-savings months |
| Any segment, secured by collateral | ceiling +10 points | safe factor +15 points, hard-capped by LTV |

The disposable-income method is the whole answer to “why are the two O2 numbers different”: a lender's basic FOIR check usually looks only at income vs. formal EMIs and ignores rent and lifestyle cost. The borrower can't ignore those. That gap is real, not a rounding difference — say so on the card.

### Pricing

| Known score | Unsecured band |
|---|---|
| ≥750 | 10.5–13% |
| 700–749 | 13–16% |
| 650–699 | 16–20% |
| <650 | 20–24% — flag likely decline unsecured |

**Unknown score:** never assume a value — not 300, not an average. Use the 650–699 band widened ±2 points and say plainly that a bureau check would likely tighten it. **Collateral present and within LTV** (property 60%, gold 75%): route to a secured product priced off the collateral (LAP 9.5–12.5%, gold loan 9–13%), score becomes irrelevant to eligibility. **APR** = quoted rate + (processing fee % ÷ tenure years) — a simplified straight-line estimate; say plainly it isn't the exact RBI Key Fact Statement XIRR method.

### Verdict logic (O1)

- A payment **bounce in the last 6 months** (informal segment) → **Don't borrow**, full stop, regardless of what the other numbers say.
- Existing EMI ≥ 30% of income *and* it includes flagged high-cost debt (24%+, e.g. app loans) → **Don't borrow.**

- Requested amount exceeds the safe-carry ceiling, no red flags → **Borrow less**, recommend the safe-carry number.
- Otherwise → **Borrow.** If the ask sits well under the safe ceiling, say so — the real constraint may be rate, not amount.

## 4. Three worked examples (numbers from the attached POC)

These ran through the actual rules engine in `borrower_copilot_poc.html` — open it and change any input to see the numbers move. Yours don't need to match; the brief says so explicitly. What has to match is the *shape* of the reasoning.

### Priya — 29, Bengaluru, salaried, Rs 1,10,000/mo, score 780

| Output | Result |
|---|---|
| O1 | **Borrow.** Rs 8,00,000 sits well inside her safe capacity. |
| O2 | Lender-likely ≈ Rs 17.3L · Safe-carry ≈ Rs 12.7L (both far above her ask) |
| O3 | 10.5–13%, all-in APR ≈ 13.5% (2% processing fee over 4yr tenure) |
| O4 | Safe EMI ceiling ≈ Rs 34,000/mo; her actual EMI on Rs 8L is ≈Rs 21,500 — comfortable even at a 20% income drop or +2pt rate. |

The insight worth putting on her card: **affordability isn't her constraint — rate is.** Her negotiating leverage should go toward pushing the lender toward 10.5% instead of accepting a generic 14% personal-loan quote, not toward asking for more money.

### Ravi — 42, Mysuru, self-employed 14yr, no formal credit history, owns his shop (Rs 45L)

| Output | Result |
|---|---|
| O1 | **Borrow less** than his Rs 15L ask, on income grounds — with a named path to close the gap. |
| O2 | Routed to LAP. Lender-view ≈ Rs 18.8L, safe-carry ≈ Rs 4.4L on his own (haircut, unverified cash) income; LTV cap ≈ Rs 27L doesn't bind — income does. |
| O3 | 9.5–12.5%, priced off the shop, not his (nonexistent) credit score. |
| O4 | The gap between Rs 4.4L safe and Rs 15L wanted is real — the card should say exactly what closes it. |

> **The scored moment:** an unsecured business loan for Ravi is the single most common wrong answer. He has no bureau file and volatile cash income — on paper he looks weak. He also owns unencumbered property worth 3x his ask. Routing him to LAP is the whole point of the exercise.

Make the card actionable, not just a number: show him three concrete levers — add his wife as a formal co-applicant, produce 6 months of bank statements to cut the cash-income haircut from 40% to 15%, or take a smaller amount now and a top-up after a year of on-time repayment history.

### Anita — 35, Hubballi, informal, existing 30%+ debt, bounce last month

| Output | Result |
|---|---|
| O1 | **Don't borrow.** The bounce alone is a hard stop under the rule above. |
| Why | Her disposable income (income – existing EMI – household expenses) already computes to zero before any new debt is added — the numbers agree with the bounce, they don't just decorate it. |

| Output | Result |
|---|---|
| Card | Not a rate to negotiate. Reframe to: stop new borrowing, get current EMIs current for 1–2 cycles, consider a small secured gold loan (if any jewellery exists) to consolidate the highest-rate debt instead of a fourth unsecured line. |

> **The scored moment:** most naive calculators will still hand her a number because “how much can she afford” always returns something. The brief calls this out by name — “Don't borrow” must be a real, reachable answer, and Anita is the test case for it. A Negotiation Card that pivots to “what to do instead” for her is a strong, differentiating signal.

## 5. Question design — what each question is for

Must-tier (8–10, everyone answers): purpose, amount wanted, loan type, employment type, income (form varies by type), existing EMI total, household expenses, age, credit score or “don't know.” Everything past this point is additional — and every row below exists only because it moves one of the four outputs. If a question you add doesn't change a number, the brief tells you to cut it, and means it.

| Segment | Additional question | What it moves |
|---|---|---|
| Self-employed | Business vintage | Under 3yr → falls back to the stricter informal FOIR ceiling (O2) |
| Self-employed | Bank statements available? | Cash-income haircut: 40% → 15% (O2) |
| Self-employed | Collateral type & value | Unlocks secured routing, better rate band, higher ceiling (O2, O3) |
| Self-employed | Co-applicant income | Adds to effective income (O2, O4) |
| Informal | Recent payment bounce? | Hard “Don't borrow” trigger (O1) |
| Informal | Existing high-cost debt? | Feeds the “Don't borrow” FOIR+flag trigger (O1) |
| Informal | Emergency savings, months | ±10pt adjustment to the safe-carry ceiling (O2, O4) |
| All | Credit score known? | Known → tight band. Unknown → widened band, never a guessed default (O3) |

## 6. Where the 100 points actually are

| Area | Points | Effort this deserves |
|---|---|---|
| Domain reasoning | 30 | Most of your 12–16 hours. This is Sections 3–4 above, made explicit in code and RULES.md. |
| Question design | 20 | A tight must-set + a handful of additional questions you can each defend in one sentence. |
| Explainability & Card | 20 | Every number renders its “why” inline, from the same variables that computed it — never a separately-written explanation that could drift from the logic. |
| Product craft | 15 | Clean, honest, mobile-first. Ranges shown as ranges. Confidence shown honestly. Not a design system. |

| Area | Points | Effort this deserves |
|---|---|---|
| Engineering | 10 | Rules in one place, UI in another, runs first time from the README. |
| Honesty about limits | 5 | RULES.md says outright what you don't know; cheap to get, easy to forget. |

Read as effort allocation: **70% of the score is reasoning and its explanation** (domain reasoning + question design + explainability). Product craft and engineering hygiene matter but are a quarter of the weight combined. If you're tempted to spend hour eight on a nicer progress bar instead of the Ravi routing logic, don't.

## 7. Architecture recommendation

- **Next.js + TypeScript** (or plain Vite + React + TS) — client-side only, no backend, matches "no login, no bureau pull, no data stored" directly instead of just promising it in the README.
- **Rules physically separate from UI**: a `/lib/rules/config.ts` (pure data — every threshold in this brief lives here) and `/lib/rules/engine.ts` (pure functions, no React, fully unit-testable). Components only call the engine and render what it returns.
- **RULES.md mirrors config.ts.** If a threshold changes, both change in the same commit — this is what "rules separated from UI, readable" is actually checking for.
- **Unit-test the engine against the three personas** as fixtures (Vitest/Jest). Given your Spotter project already used this pattern (PostGIS/Redis service with 49 unit tests), this should be familiar territory — reuse that instinct here.
- **Negotiation Card** exports client-side (print-to-PDF or a lightweight lib like jsPDF) — no server round-trip needed for a one-page card.
- **No ML, no bureau API, no persistence.** The brief says these aren't scored; building them anyway makes your reasoning harder to audit, which actively works against the explainability score.

## 8. Traps most other candidates will fall into

- Unknown credit score quietly defaulted to some fixed number instead of genuinely widening the range (the brief calls this out by name: "unknown is never zero").
- Ravi priced as an unsecured self-employed borrower — ignoring the collateral sitting right in his profile.
- A calculator that *can't* say "don't borrow" — optimistic by construction, which fails Anita's case specifically.
- O2 collapsed into one number instead of two — losing the lender-vs-borrower distinction that's worth 30 points on its own.
- APR that's just the headline rate re-labelled, with no fee folded in.
- One long linear questionnaire, ignoring the adaptive/skip-logic requirement — a kirana owner shouldn't see salaried-employee questions.
- RULES.md written as prose instead of the exact table format asked for: *what · value · why · source or "my judgement."*
- Rules logic embedded inside UI components — fails the engineering point even if the numbers are right.

## 9. Running the four days

| When | Do |
|---|---|
| Day 1 | Lock the rules framework (Sections 3–4) and write RULES.md first, before any UI. Run the three personas through it by hand or in a scratch script. |
| Day 2 | Build the engine as pure functions + config, with unit tests against the three personas. This is the highest-value 8 hours of the whole challenge. |
| Day 3 | Adaptive question flow + the four output cards + Negotiation Card, wired to the engine you already trust because it's tested. |
| Day 4 | Polish copy, mobile check, write the three run-throughs and the five-minute walkthrough. Reread RULES.md and make sure you could defend every row of it out loud, live, with no notes. |

That last line matters more than it looks: the follow-up is 60 minutes, one persona walked through, then they change an assumption and watch you change the app. If a number in your app can't be traced to a line in RULES.md you personally understand, that's the moment it shows.

## 10. What comes with this brief

- **BUILD_PROMPT.md** — a paste-ready master prompt for Claude Code / Cursor / any AI coding agent, encoding everything above (rules, question tree, architecture, personas, deliverables) so the scaffolding gets built fast and you spend your hours on judgement, not boilerplate.
- **borrower_copilot_poc.html** — a working, single-file proof of concept: the rules engine above, actually running, with all three personas selectable and every "why" rendered inline. Open it in any browser, no install. Read the <script> tag — every threshold in this brief is a named constant in there, not buried in a formula.

> **One condition, from their own brief:** "you must be able to defend every rule without them." Use both files to move fast, then sit with RULES.md until every number in it is something you'd say the same way twice, unprompted, a week from now.

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*What they're really testing: can lending judgement become rules a borrower can see and a machine can run. That's also, not coincidentally, what this brief just tried to model for you.*